

SHASHANK BHAGAVATULA

309-622-2695 | sbhagavatula03@gmail.com | github.com/sbhag3

12650 120th Ave NE Unit 529, Kirkland, WA, 98034

EDUCATION

University of Illinois at Urbana-Champaign

Professional Masters of Computer Science in Data Science and AI (Part-time)

Urbana-Champaign, IL

Graduating Aug 2027

University of Illinois at Urbana-Champaign

Bachelor of Science in Mathematics and Computer Science

Urbana-Champaign, IL

TECHNICAL SKILLS

Languages: Python, Java, C/C++, GoLang, SQL/NoSQL, Terraform, JavaScript, TypeScript, HTML/CSS, R, Haskell

Frameworks: React, Node.js, Flask, JUnit, WordPress, Fastify, Django, Spring Boot, Neo4j, d3

Developer Tools: Git, Docker, Amazon Web Services (AWS), PostgreSQL, Visual Studio Code, PyCharm, IntelliJ, Eclipse, Tableau, Claude Code, Codex, Gemini

Libraries: pandas, NumPy, Matplotlib, TensorFlow, sklearn, PyTorch, SQLite3

Certifications: AWS Certified Cloud Practitioner (CCP), Hashicorp Terraform Associate

EXPERIENCE

Software Engineer

Aug 2025 - Present

Google

Kirkland, WA

- Led the design and implementation of a two-tier, multi-replica shuffle sharding architecture for Slicer's assignment distributor, enhancing service resilience and load distribution.
- Improved Slicer data propagation efficiency by 50% by optimizing Slicer's data sharding assignment application logic, with improvement validated using detailed and automated Java microbenchmarks and metric implementation.
- Strengthened the reliability of Slicer's production monitoring by refining proxy alerts, reducing false positive rate by 20% after filtering for supported cells.

Software Engineer

Jan 2023 – May 2025

State Farm

Atlanta, GA

- Developed a data dashboard to allow claims data analysts to further their statistical research by using various forms of data visualizations and graphs for newly established claims and policies.
- Built a data pipeline to extract and reconcile claims data between the Index-local database and the system-of-record source of truth, surfacing discrepancies through an analytics dashboard for the Agreement Index team.
- Built an ETL pipeline to migrate Life Communications' notice-letter data from legacy Lotus Notes to AWS, extracting and transforming legacy records into a cloud-native format and reducing automated letter-generation write time by 33%.
- Designed an ETL pipeline ingesting relational claims and participant data into a Neo4j graph, transforming records into nodes and edges to enable Graph ML fraud detection and reducing data querying by nearly 80%.
- Developed infrastructure for several APIs in Java and GoLang to manage rental/repair eligibility and coverage options for claimholders while moving to a cloud solution, improving service efficiency.

Undergraduate Course Assistant

University of Illinois at Urbana-Champaign

Champaign-Urbana, IL

- Responsible for assisting with teaching CS 233 (Computer Architecture) to over 700 students with weekly office hours and lab assignment assistance.

Infrastructure Team Co-Lead

ACM@UIUC

Champaign-Urbana, IL

- Led development on a management API integrated with Azure AD and Discord API to allow ACM@UIUC's leads to assign leadership roles on major communication platforms. Moved important TypeScript services - link shortener, Stripe link generator for company sponsors, etc. - to ACM@UIUC's Core API.

PROJECTS

GreenChain Insights Dashboard | Python, SQL, React, d3, Tableau

Feb 2025 – May 2025

- Built an ETL backend ingesting and structuring GHG emissions data across 1000+ industries, optimizing queries to reduce data retrieval by 68% and powering an interactive d3/React dashboard.
- Developed "Explore Carbon Footprint" tool to simulate purchase impacts; optimized queries, reducing data retrieval by 68%.

Agenda Visualization Application (AVA) | Python, C++, HTML/CSS, JavaScript

Aug 2024 – Dec 2024

- Created tool to help students optimize task scheduling, reducing planning time; submitted to HackIllinois 2024.
- Enhanced with TensorFlow ML model, LLM-based suggestions, and login/save features.

Malware Detection Classifier | VirusTotal API, Python, HTML/CSS, JavaScript

Aug 2023 – Dec 2023

- Developed tool leveraging VirusTotal API to identify system threats; submitted to HackIllinois 2023.
- Built Python ML model to handle complex prompts and predict recurring malware risks.